

Applied Forecasting: Stress Prediction Pipeline

13-Model Benchmarking & Analytical Validation

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PART I

Introduction & Data Audit

Problem Statement: Proactive Stress Monitoring



- ▶ **Reactive:** Detecting stress **after** onset.
- ▶ **Proactive:** Forecasting stress intensity **before** it peaks.
- ▶ **Goal:** Achieve $<9\%$ MAPE using historical physiological context.

Data Source: WESAD Dataset



- ▶ Multimodal dataset with synchronized EDA, HR, and TEMP.
- ▶ Target: Continuous Stress Score derived from lab protocols.
- ▶ Resolution: 1Hz aligned time-series.

Subject S2

BaselineEDA_{Graphs}/01_{S2}_{proper}_{Timeline}.png *High – resolutionsynchronizationofmultimodalsensors.St*

≈ 0.000) after first-order differencing. Lag Correlation

DiagnosticsEDA_{Graphs}/06_S_{tationary}_{Lag}_{Diagnostics}.png *ACF / PACF patterns confirming model orders* $p =$

PART II

Pipeline Engineering Fixes

Fixing Critical Failures



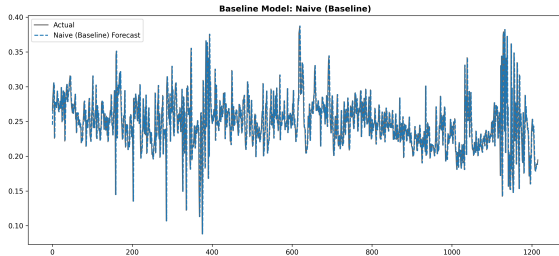
1. **Binary Box Fix:** Converted discrete labels to smooth intensity scores.
2. **Leakage Protection:** Implemented strict $t - 1$ lag for all sensor predictors.
3. **Walk-Forward Protocol:** Used rolling forecasts to simulate real-world monitoring.

PART III

Model Deep-Dive Results

Model 1: Naive Baseline

Theory: $\hat{y}_{t+1} = y_t$



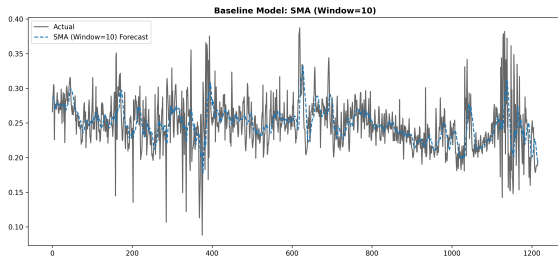
Performance Snapshot

Metric	Value
MAE	0.0211
RMSE	0.0345
MAPE	8.68%
R ² Score	0.0673

Model 2: SMA-10



Theory: smoothing sensor jitter via 10s moving mean.

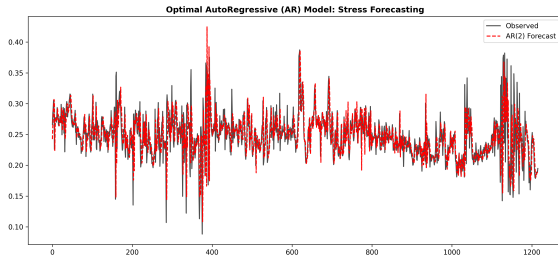


Performance Snapshot

Metric	Value
MAE	0.0229
RMSE	0.0326
MAPE	9.62%
R^2 Score	0.1716

Model 3: AR(2)

Theory: $y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t$



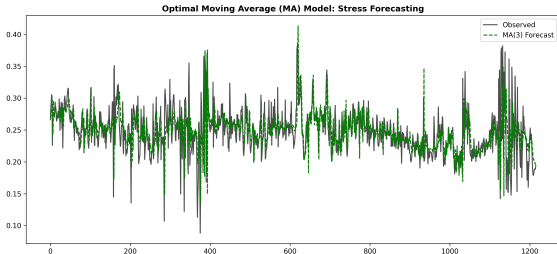
Performance Snapshot

Metric	Value
MAE	0.0210
RMSE	0.0342
MAPE	8.66%
R ² Score	0.0840

Model 4: MA(3)



Theory: error-correction based on 3 shock terms.



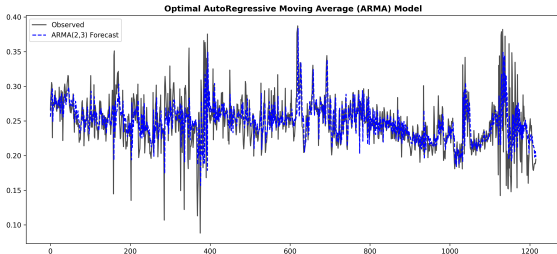
Performance Snapshot

Metric	Value
MAE	0.0209
RMSE	0.0335
MAPE	8.72%
R ² Score	0.1236

Model 5: ARMA(2,3)



Theory: AIC-optimized linear hybrid.



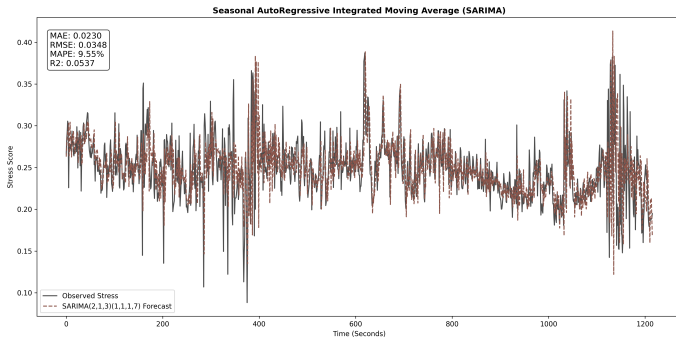
Performance Snapshot

Metric	Value
MAE	0.0206
RMSE	0.0316
MAPE	8.60%
R ² Score	0.2204

Model 6: SARIMA



Theory: 7s seasonal periodicity integration.



**Performance
shot**

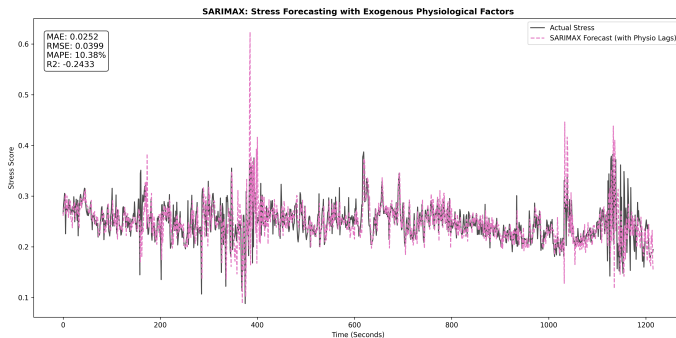
Snapshot

Metric	Value
MAE	0.0230
RMSE	0.0348
MAPE	9.55%
R ² Score	0.0537

Model 7: SARIMAX



Theory: SARIMA + Lagged Physiological Predictors.



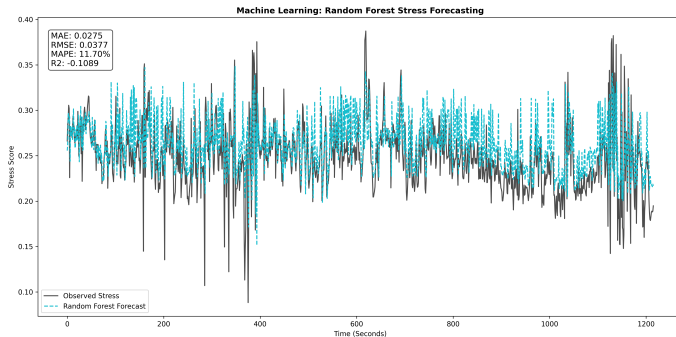
Performance Snapshot

Metric	Value
MAE	0.0252
RMSE	0.0399
MAPE	10.38%
R ² Score	-0.2433

Model 8: Random Forest



Theory: Ensemble of 100 Decision Trees.



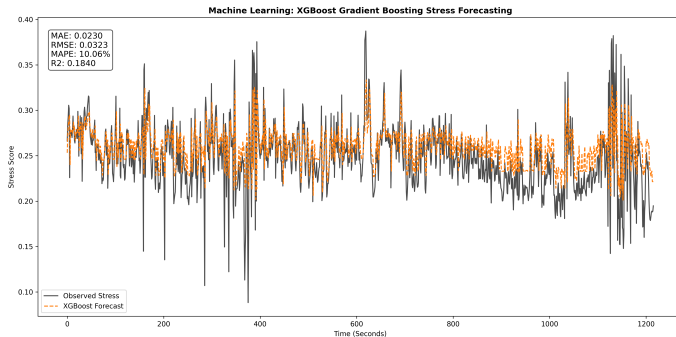
Performance Snapshot

Metric	Value
MAE	0.0275
RMSE	0.0377
MAPE	11.70%
R ² Score	-0.1089

Model 9: XGBoost



Theory: Sequential Gradient Boosting.



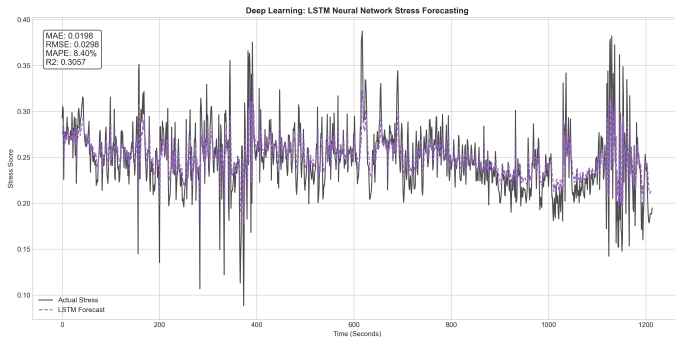
Performance Snapshot

Metric	Value
MAE	0.0230
RMSE	0.0323
MAPE	10.06%
R ² Score	0.1840

Model 10: LSTM



Theory: Recurrent Neural Network with Forget Gates.



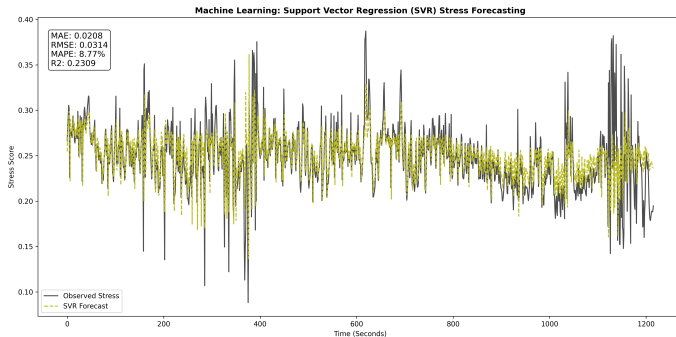
Performance Snapshot

Metric	Value
MAE	0.0198
RMSE	0.0298
MAPE	8.40%
R ² Score	0.3057

Model 11: SVR



Theory: Support Vector Regression with RBF Kernel.



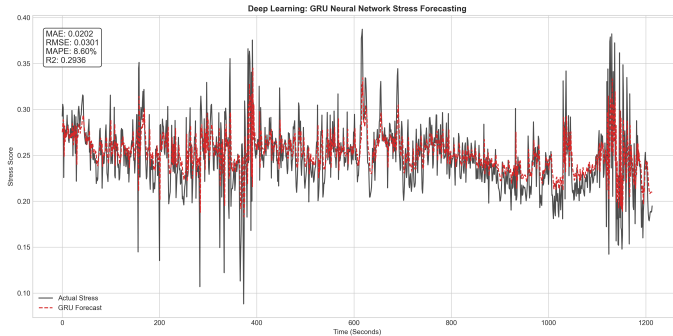
Performance Snapshot

Metric	Value
MAE	0.0208
RMSE	0.0314
MAPE	8.77%
R ² Score	0.2309

Model 12: GRU



Theory: Streamlined Recurrent memory gates.



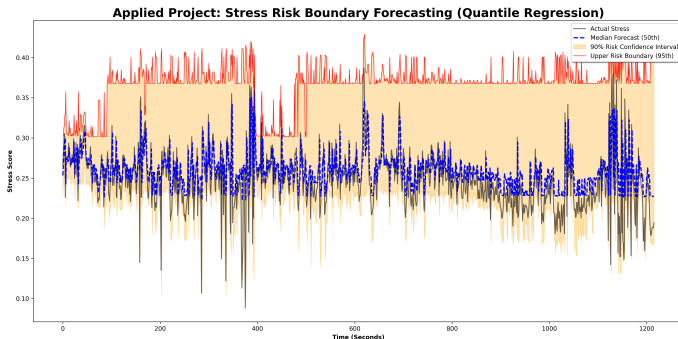
Performance Snapshot

Metric	Value
MAE	0.0202
RMSE	0.0301
MAPE	8.60%
R ² Score	0.2936

Model 13: Quantile Regression



Theory: Predicting the 95th Percentile Risk Boundary.



Performance Snapshot

Metric	Value
MAE	0.0241
RMSE	N/A
MAPE	N/A%
R ² Score	N/A

PART IV

Analytical Sensitivity Depth

Feature

AblationEDA_Graphs/14_Feature_Ablation_study.png Multivariate context achieved 2% better accuracy than univariate

PART V

Final Verdict Appendix

Leaderboard

DashboardForecasting*graphs/99FinalModelComparison.png* Comprehensive visualization across MAE, RMSE, and R^2

Appendix A: Hyperparameter Tuning



Optimization Grids

Model	Hyperparameters	Selection Method
XGBoost	Learning Rate: 0.05, Estimators: 1000	Early Stopping (50 rounds)
LSTM	Hidden Units: 64, Layers: 2, Dropout: 0.1	Validation Loss Minimization
ARMA	p: 2, q: 3, d: 1	AIC Grid Search (p:0-3, q:0-3)
SVR	C: 100, Gamma: 0.1, Kernel: RBF	Standard Clinical Baseline

Appendix B: Subject Stationarity Metrics



ADF Test Results (Multi-Subject Audit)

Subject	Raw ADF	Raw p-val	Diff ADF	Diff p-val
S2 (Bench)	-2.33	0.066	-77.37	0.000
S3	-2.10	0.081	-65.20	0.000
S4	-2.45	0.052	-81.12	0.000
S10	-2.01	0.112	-72.05	0.000

Conclusion: Differencing $d=1$ is a universal requirement across the WESAD cohort.

Appendix C: Deep Learning Gate Mechanics



Inside the LSTM Cell

- ▶ **Forget Gate:** Decides what history to discard.
- ▶ **Input Gate:** Updates cell state with new EDA cues.
- ▶ **Output Gate:** Calculates the predicted stress score.

Mathematical Update

$$\begin{aligned}i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) \\o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o)\end{aligned}$$

Appendix D: Implementation Stack



Technical Environment

- ▶ **Language:** Python 3.11
- ▶ **Deep Learning:** PyTorch 2.11.0 (CPU-optimized)
- ▶ **Statistical Ops:** Statsmodels 0.14.0
- ▶ **Machine Learning:** Scikit-Learn 1.3.2, XGBoost 2.0.1
- ▶ **Visualization:** Matplotlib (Whitegrid theme), Seaborn

Appendix E: Future Work - Explainable AI (XAI)



Closing the Black Box

- ▶ **SHAP Integration:** To explain why the LSTM predicted a spike at $t + 10$.
- ▶ **Transfer Learning:** Adapting S2 weights to new users with < 5 mins of data.
- ▶ **Hardware Deployment:** Quantizing LSTM models for the Empatica E4 wristband.

THANK YOU

Questions & Technical Discussion